

UNIT 2: PRACTICE QUESTION BANK & ANSWER KEY

Course Pattern: TE IT - 2024 Pattern (SPPU)

Academic Term: Semester V (2026)

Document Type: Question Bank with Answer Key

Course Code: PCC-301-ITT

Section A: Important Theory Questions

- Process Description and Control Block:**
 - Explain the concept of a process. Draw and explain the 7-State Process Model with state transition details. [SPPU Nov 2023, Marks: 8]
 - Describe the structure and attributes of a Process Control Block (PCB). Explain why the PCB is critical during context switching. [SPPU May 2024, Marks: 6]
- Process Control API & Multithreading:**
 - Differentiate between a Mode Switch and a Context Switch with respect to execution overhead and CPU register saving. [SPPU Dec 2022, Marks: 6]
 - Explain the `fork()`, `exec()`, and `wait()` system calls. Differentiate between a Zombie Process and an Orphan Process. [SPPU Nov 2023, Marks: 8]
 - Compare User-Level Threads (ULT) and Kernel-Level Threads (KLT). What happens in ULT when a thread makes a blocking I/O call? [SPPU May 2023, Marks: 6]
- CPU Scheduling Principles:**
 - Define CPU Scheduling criteria: Throughput, Turnaround Time, Waiting Time, Response Time, and CPU Utilization. [SPPU May 2024, Marks: 6]
 - Explain Multilevel Feedback Queue (MLFQ) scheduling and describe how it prevents starvation. [SPPU Dec 2023, Marks: 6]

Section B: Numerical CPU Scheduling Practice Problems

Practice Problem 1: FCFS and SJF Scheduling

Consider the following set of processes:

Process	Arrival Time (T_A ms)	Burst Time (T_B ms)
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

Draw Gantt charts and calculate Average Turnaround Time (ATAT) and Average Waiting Time (AWT) using:

- (a) FCFS Scheduling [Marks: 4]
- (b) Non-Preemptive SJF Scheduling [Marks: 4]

Practice Problem 2: Preemptive SRTF Scheduling

Consider the process table:

Process	Arrival Time (T_A ms)	Burst Time (T_B ms)
P_1	0	6
P_2	1	3
P_3	2	1
P_4	3	4

Draw the SRTF Gantt chart and compute Completion Time, Turnaround Time, and Waiting Time for each process. Find ATAT and AWT. [SPPU Nov 2023, Marks: 8]

Practice Problem 3: Round Robin Scheduling ($q = 3$ ms)

Given the processes below:

Process	Arrival Time (T_A ms)	Burst Time (T_B ms)
P_1	0	7
P_2	2	4
P_3	4	2

Construct the Round Robin Gantt chart for Time Quantum $q = 3$ ms and calculate ATAT and AWT. [SPPU May 2024, Marks: 8]

Mandatory Final Answer Key

Below are the verified numerical answers for verification:

• Practice Problem 1 (a) — FCFS:

- Gantt Chart Execution: $P_1(0 \rightarrow 8), P_2(8 \rightarrow 12), P_3(12 \rightarrow 21), P_4(21 \rightarrow 26)$
- Completion Times: $P_1 = 8, P_2 = 12, P_3 = 21, P_4 = 26$
- Turnaround Times: $P_1 = 8, P_2 = 11, P_3 = 19, P_4 = 23$
- Waiting Times: $P_1 = 0, P_2 = 7, P_3 = 10, P_4 = 18$
- **ATAT = 15.25 ms, AWT = 8.75 ms**

• Practice Problem 1 (b) — Non-Preemptive SJF:

- Gantt Chart Execution: $P_1(0 \rightarrow 8), P_2(8 \rightarrow 12), P_4(12 \rightarrow 17), P_3(17 \rightarrow 26)$
- Completion Times: $P_1 = 8, P_2 = 12, P_4 = 17, P_3 = 26$
- Turnaround Times: $P_1 = 8, P_2 = 11, P_4 = 14, P_3 = 24$
- Waiting Times: $P_1 = 0, P_2 = 7, P_4 = 9, P_3 = 15$
- **ATAT = 14.25 ms, AWT = 7.75 ms**

• Practice Problem 2 — SRTF:

- Gantt Chart Execution: $P_1(0 \rightarrow 1), P_2(1 \rightarrow 2), P_3(2 \rightarrow 3), P_2(3 \rightarrow 5), P_4(5 \rightarrow 9), P_1(9 \rightarrow 14)$
- Completion Times: $P_1 = 14, P_2 = 5, P_3 = 3, P_4 = 9$
- Turnaround Times: $P_1 = 14, P_2 = 4, P_3 = 1, P_4 = 6$
- Waiting Times: $P_1 = 8, P_2 = 1, P_3 = 0, P_4 = 2$
- **ATAT = 6.25 ms, AWT = 2.75 ms**

• Practice Problem 3 — Round Robin ($q = 3$ ms):

- Gantt Chart Execution: $P_1(0 \rightarrow 3), P_2(3 \rightarrow 6), P_1(6 \rightarrow 9), P_3(9 \rightarrow 11), P_2(11 \rightarrow 12), P_1(12 \rightarrow 13)$

- Completion Times: $P_1 = 13, P_2 = 12, P_3 = 11$
 - Turnaround Times: $P_1 = 13, P_2 = 10, P_3 = 7$
 - Waiting Times: $P_1 = 6, P_2 = 6, P_3 = 5$
 - **ATAT = 10.00 ms, AWT = 5.67 ms**
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End of Operating Systems Unit 2 Practice Question Bank & Answer Key — SPPU TE IT 2024 Pattern